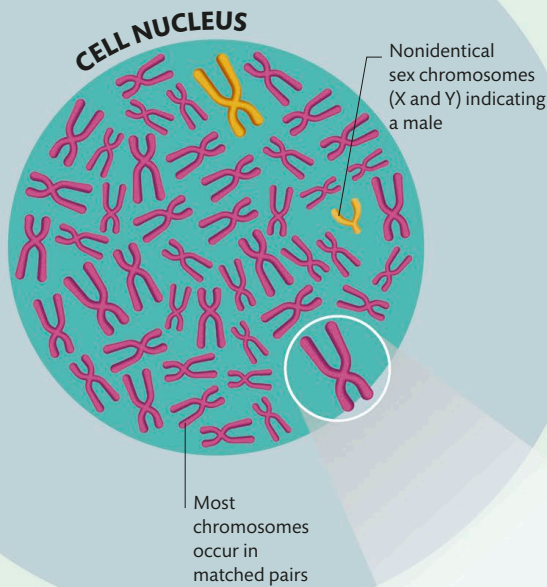




Chromosomes

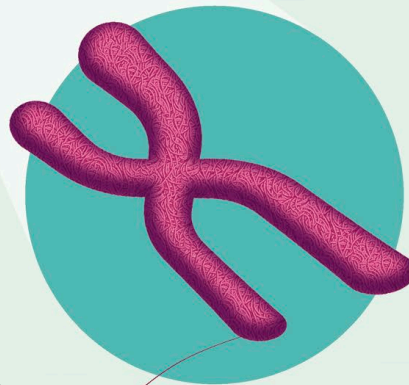
We have around 20,000 genes, which are grouped into chromosomes. Each cell nucleus has 22 matched pairs of chromosomes (known as autosomes), plus a pair of sex chromosomes (identical XX chromosomes in females, or a nonidentical pair, XY, in males).

ARE GENES ALWAYS ACTIVE?

Every DNA-bearing cell has a full set of genes, but many genes are normally active in only one part of the body, such as the brain, or at one stage of life, such as babyhood.

What is a gene?

Genes are sections of a long molecule called deoxyribonucleic acid (DNA), which contains the code that governs how our bodies develop and function. We inherit a mixture of genes from our parents. These genes produce proteins that shape physical traits, such as eye color, or regulate processes such as chemical reactions. Their action turns these traits “on” or “off” or makes them more or less intense.

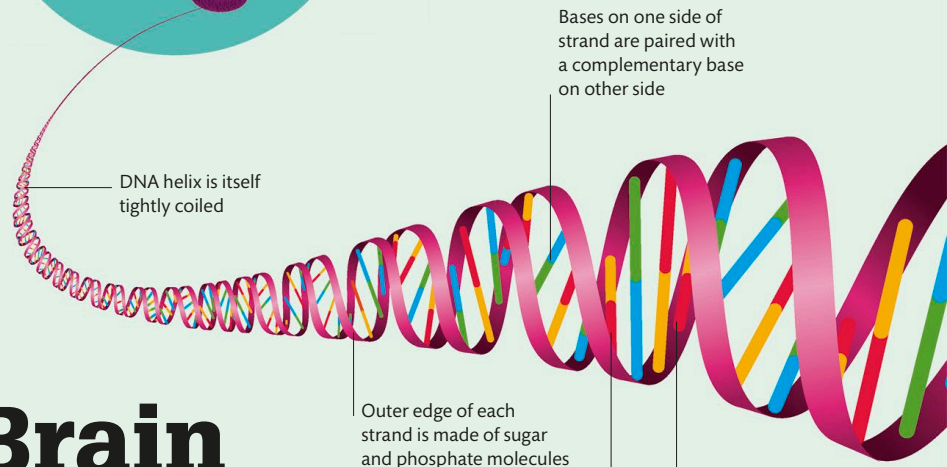


DNA helix is itself tightly coiled

DNA and genes

The DNA molecule is a long, twisted strand formed from pairs of chemicals called bases—the “letters” of the genetic code—with a sugar-phosphate backbone at each edge. When cells divide, half of the DNA goes into each new cell. In addition, we inherit one chromosome in each pair from our mother and one from our father, so each parent contributes half of our genes.

Bases on one side of strand are paired with a complementary base on other side



Outer edge of each strand is made of sugar and phosphate molecules

Four bases—adenine, thymine, guanine, and cytosine—are arranged in a particular sequence that encodes our genetic information

Adenine (red) always bonds with thymine (yellow)

Genetics and the Brain

Genes govern the way our bodies, including the brain, develop and function. They work together with our environment to shape us throughout our life, from conception to old age.